

The promise of quantum computers is that certain computational tasks might be executed exponentially faster on a quantum processor than on a classical computer. We demonstrate quantum supremacy using a programmable superconducting processor named Sycamore.

Our Sycamore processor consists of a two-dimensional array of 54 transmon qubits, where each qubit is capacitively coupled to four nearest neighbors in a square lattice geometry.

Cross-entropy benchmarking fidelity: We assess the fidelity of our quantum circuits by comparing the probability distribution of measured bitstrings with the simulated classical probabilities.

The Sycamore processor took approximately 200 seconds to sample one instance of a quantum circuit a million times, whereas a state-of-the-art classical supercomputer would require 10,000 years for the same task.

Quantum error correction will be required to achieve fault-tolerant general-purpose quantum computing. Surface codes with physical error rates below threshold are demonstrated.

Calibration of single-qubit microwave drive pulses and two-qubit couplers is performed using randomized benchmarking and continuous parameter optimization.